INCH-POUND

MIL-PRF-49149D(CR)
4 March 1999
SUPERSEDING
MIL-E-49149C(CR)
27 May 1992

PERFORMANCE SPECIFICATION

ELECTRONIC CONTROL ASSEMBLY
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense,

1. SCOPE

1.1 Scope. This specification covers the Electronic Control Assembly herein referred to as electronic control, which is part of Viewer, Driver's Night Vision AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.1.1 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.d).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771336 - Electronic Control Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The electronic control is mounted to and limits operation of the Viewer, Driver's Night Vision AN/VVS-2(V) under damaging light brightness level.

3.2 Construction. The electronic control shall be constructed to assure compliance with the contract configuration baseline and in accordance with SM-D-771336 as applicable.

3.3 First article testing (FAT). When specified in the contract (see 6.2.c and 6.2.f) the contractor shall furnish units for FAT inspection and approval (see 4.2).

3.4 Materials. The contractor shall select parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Performance characteristics.

NOTE: The performance tests of 3.5 shall be performed with a 51 ±2 ohm load between pins 1 and 4 of the electronic control.

3.5.1 Power output. When the photo-detector is blocked, the electronic control shall be within ±0.3 Vdc of the input voltage when energized with a BA-5567/U battery, or equivalent DC power source.

3.5.2 Power cutoff. The output voltage of the electronic control shall be reduced to 0.2 Vdc or less when the photodetector views a bright target area between 0.1 and 0.18 footlamberts.

3.5.3 Power restoration. Output voltage shall return to within + 0.2

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Vdc of the input voltage within 5 seconds after the brightness of the target area is reduced below the cutoff value of the photodetector.

3.6 Environmental characteristics.

3.6.1 Temperature Extremes. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being operated at a temperature of +52°C or -51°C. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to storage at +68°C or -51°C. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to repetitive temperature changes between +23°C and -51°C within 5 minutes and +23°C and +68°C within 5 minutes. The soak time at temperature extremes shall be sufficient for the entire electronic control to achieve stabilization a the specified temperature. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature. Operation of the electronic control is not required during storage testing.

3.6.2 Vibration. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation. Vibration shall be limited to one sine sweep from 5.0 to 500 Hertz (Hz) and back to 5.0 Hz within 15 minutes in the vertical axis only. Vibration shall be performed from 5.0 to 5.5 Hz at 1 inch displacement, 5.5 to 30 Hz. at 1.5 g’s peak, 30 to 50 Hz. at 0.33 inch displacement and 50 to 500 Hz. at 4.2 g’s peak

3.6.3 Shock.

3.6.3.1 Basic shock. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to basic shock of 40 g's (see 3.10.5) for a duration of 18 milliseconds.

3.6.3.2 Weapon shock. The electronic control shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to simulated weapon firing shock of 100 g's (see 3.10.5) at 1.5 milliseconds on half sine pulse.

3.7 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract or purchase order.

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3.8 Treatment and painting. Treatment and painting shall be in accordance with applicable drawings unless otherwise specified in the contract or purchase order.

3.9 Workmanship. Standards of workmanship shall be such that the electronic control will meet all requirements of this specification and any reference specifications or drawings.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as 23°C, ± 10°C.

3.10.3 Environmental temperature. Environmental chamber temperatures shall be controlled to within ±5°C of the specified temperature unless otherwise specified.

3.10.4 Degradation of performance. Degradation of performance is defined as failure to operate in accordance with the performance requirements of this specification.

3.10.5 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be as classified as follows:

a. First article tests (4.2).
b. Conformance inspection (4.3).

4.2 First article testing (FAT). When specified in the contract or purchase order (see 6.2.f), first article testing shall be performed by the contractor. The quantity of first article test samples shall be as specified in the contract or purchase order (see 6.2.c) and shall be selected from the first production lot.

4.2.1 FAT Test. Each electronic control in the first article lot shall be subjected to the performance tests in Table I. Electronic controls selected as FAT samples shall be subjected to the environmental tests of Table II. Failure of any test shall be cause for rejection of that electronic control and may be cause for rejection of the FAT.

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TABLE I. Unit first article testing.

Inspection	Requirement Paragraph
Power output	3.5.1
Power cutoff	3.5.2
Power restoration	3.5.3

TABLE II Sample first article testing.

Inspection	Requirement Paragraph
Temperature Extremes	3.6.1
Vibration	3.6.2
Shock	3.6.3

4.2.2 Disposition of FAT samples. Disposition of FAT samples shall be in accordance with the contract or purchase order (see 6.2.c).

4.3 **C**onformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.e). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.10.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the Packaging requirements shall be as specified in the contract or purchase order(see 6.2.h). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electronic control is used with the AN/VVS-2(V) for driver's viewing during periods of darkness.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Necessary actions by contractor in the event of a lot failure.
c. Quantity and schedule for FAT testing for all line items and disposition of FAT samples.
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.1.1).
e. Statement identifying that the contractor must submit his proposal for conformance inspection to include specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
f. Statement specifying that FAT testing is, or is not, required.
g. Statement identifying that the contractor must submit his proposal for Environmental Stress Screening (ESS).
h. Packaging requirements.

6.3 Definitions. See 3.10.

6.4 Subject term keyword listing.

Power cutoff
Power output
Power restoration

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49149D (CR)	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Electronic Control Assembly Viewer, Driver's Night Vision, AN/VVS-2(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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